Inference after human genetic clustering

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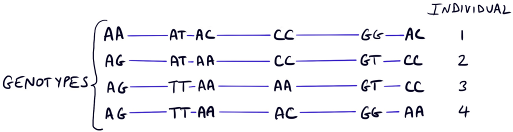
1. McGill University, 2. University of Rennes, ENSAI, CREST.

Statistical Methods for Post Genomic Data

January 29, 2026

SNP data describing genotypes in a population



Figures : *An introduction to human population genetics, variation, and disease*, Jonathan K Pritchard.

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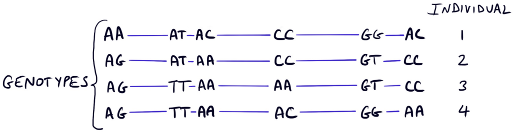

Allele configurations → SNP code

AA → 0, Aa → 1, aa → 2,
with A (resp. a) being the reference (resp. alternate) allele.

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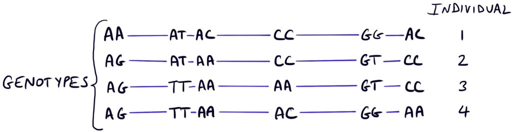

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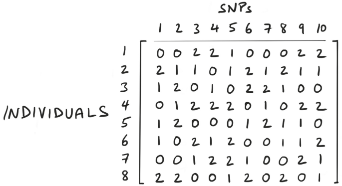

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Starting point : top 100 PCs

Example : 1KGP data

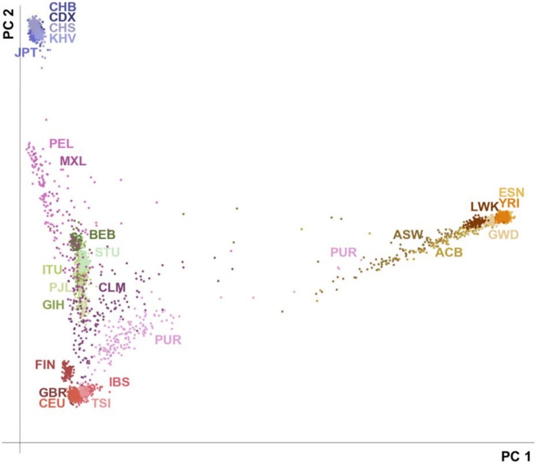

Figure : Diaz-Papkovich *et al.* (PLOS Genetics, 2017).

Population structure and genotype data

Population structure

Models to approximate a complex tangled web of past migrations and mixtures of ancestors, and nonrandom mating at different geographic scales.

$\rightarrow$ There is no single "correct" description of structure...

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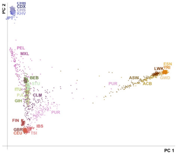
Fundamental question in Population Genetics

Can we infer/observe population structure and/or ancestry from genetic data ?

$$
\text{INDIVIDUALS}\quad
\begin{array}{c}
\\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8
\end{array}
\overset{\displaystyle\text{SNPs}}{\begin{array}{cccccccccc}
1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\
\end{array}}
$$

$$
\begin{array}{c|cccccccccc}
 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\
\hline
1 & 0 & 0 & 2 & 2 & 1 & 0 & 0 & 0 & 2 & 2 \\
2 & 2 & 1 & 1 & 0 & 1 & 2 & 1 & 2 & 1 & 1 \\
3 & 1 & 2 & 0 & 1 & 0 & 2 & 2 & 1 & 0 & 0 \\
4 & 0 & 1 & 2 & 2 & 2 & 0 & 1 & 0 & 2 & 2 \\
5 & 1 & 2 & 0 & 0 & 0 & 1 & 2 & 1 & 1 & 0 \\
6 & 1 & 0 & 2 & 1 & 2 & 0 & 0 & 1 & 1 & 2 \\
7 & 0 & 0 & 1 & 2 & 2 & 1 & 0 & 0 & 2 & 1 \\
8 & 2 & 2 & 0 & 0 & 1 & 2 & 0 & 2 & 0 & 1
\end{array}
\longrightarrow
$$



$\longrightarrow$ **Population structure**

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Can we infer/observe population structure and/or ancestry from genetic data ?

- Simple clustering algorithms like hierarchical clustering reveal broad (continental) structure[1].

1. Mountain and Cavalli-Sforza (1997).

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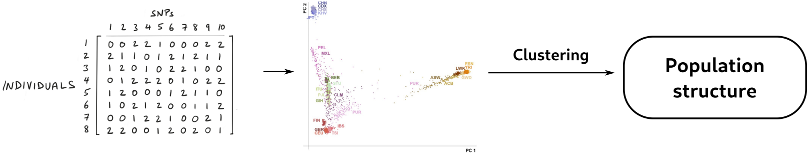

- Simple clustering algorithms like hierarchical clustering reveal broad (continental) structure[1].
- More complex algorithms can reveal population structure at finer levels[2] (e.g. the UMAP+HDBSCAN($\hat{\epsilon}$) pipeline[3]).

1. Mountain and Cavalli-Sforza (1997), 2. Rosenberg *et al.* (2002), Hollfelder *et al.* (2017), 3. Diaz-Papkovich *et al.* (2025).

Clustering genetic data

Main challenges

Clustering individuals
based on their genome

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Statistical concerns

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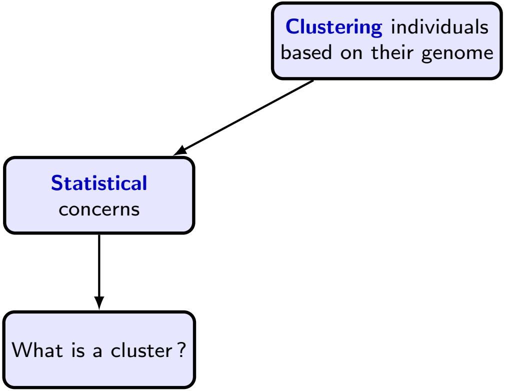
Main challenges



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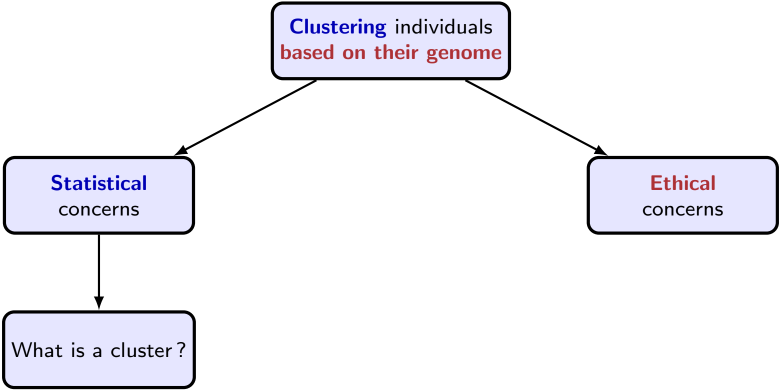

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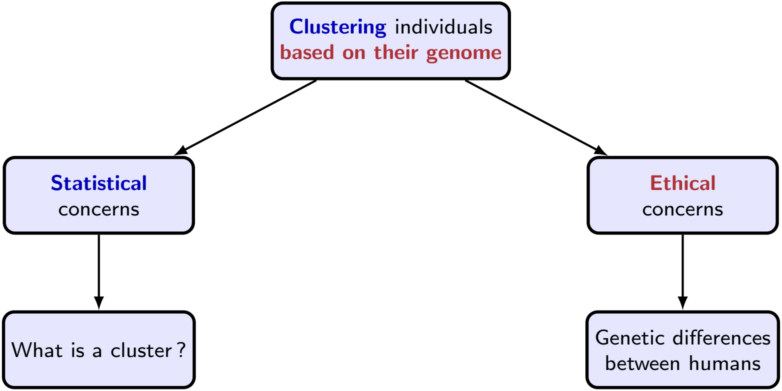

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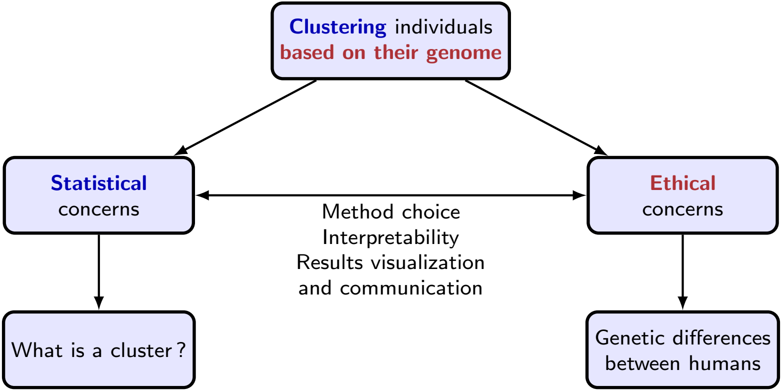

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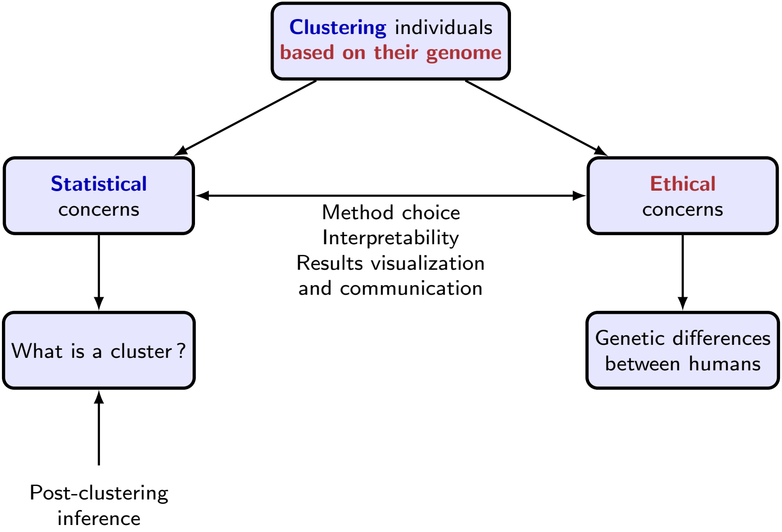

Post-clustering inference

Main idea

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- If every observation X_i has mean μ_i, the goal is to test

$$H_0^{\{\mathcal{G}_1, \mathcal{G}_2\}} : \bar{\mu}_{\mathcal{G}_1} = \bar{\mu}_{\mathcal{G}_2}, \tag{H0}$$

where $\bar{\mu}_{\mathcal{G}_k} = \frac{1}{|\mathcal{G}_k|} \sum_{i \in \mathcal{G}_k} \mu_i$, for $k \in \{1, 2\}, i \in \{1, \ldots, n\}$.

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Standard strategy : conditional inference

Control the selective type I error for clustering at level α, that is,

$$\mathbb{P}_{H_0^{\{\mathcal{G}_1, \mathcal{G}_2\}}} \left(\text{reject } H_0^{\{\mathcal{G}_1, \mathcal{G}_2\}} \text{ based on } \mathbf{X} \text{ at level } \alpha \;\middle|\; \mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}(\mathbf{X}) \right) \leq \alpha \quad \forall \alpha \in (0, 1).$$

Model assumptions in the state-of-the-art

Analitically tractable p-values require Gaussianity assumptions

- Independent observations $X_i \sim \mathcal{N}_p(\mu_i, \sigma^2 \mathbb{I}_p)$.

  $\rightarrow$ Gao *et al.* (2022), Yun and Barber (2023), Chen and Witten (2023).

- Gaussian features (feature-level test) $X_g \sim \mathcal{N}_n(\mu_g, \sigma_g^2 \mathbb{I}_n)$.

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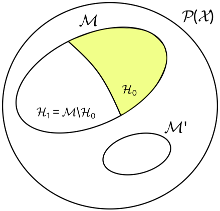
Testing when data may not fit the model

$$\text{Reject } H_0 \Longrightarrow \begin{cases} \mathcal{H}_1 : \ \bar{\mu}_{\mathcal{G}_1} \neq \bar{\mu}_{\mathcal{G}_2}, \\ \text{or} \\ \text{model misspecification.} \end{cases}$$



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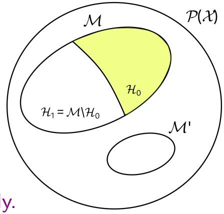

$\rightarrow$ Robustness to non-Gaussianity has been shown empirically.

Standard strategy

1. Consider a p-value of the form

$$p(x) = \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\left(\|\bar{X}_{\mathcal{G}_1} - \bar{X}_{\mathcal{G}_2}\| \geq \|\bar{x}_{\mathcal{G}_1} - \bar{x}_{\mathcal{G}_2}\| \,\middle|\, \mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}(\mathbf{X}),\ T(\mathbf{X}) \right),$$

ensuring the control of the selective type I error for clustering.

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$$p(x) = \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\left(\phi \geq \|\bar{x}_{\mathcal{G}_1} - \bar{x}_{\mathcal{G}_2}\| \,\middle|\, \mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}\left(\mathbf{x}'(\phi)\right) \right),$$

where ϕ has a known null distribution and $\mathbf{x}'(\phi)$ is a perturbation of $\mathbf{x}$

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- If $\phi \geq \|\bar{x}_{\mathcal{G}_1} - \bar{x}_{\mathcal{G}_2}\|$, $\mathcal{G}_1$ and $\mathcal{G}_2$ are pulled appart,
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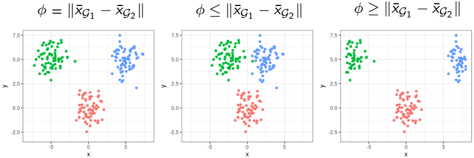

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Interpretation of p-value

A consequence of conditioning

$$
\begin{aligned}
p(x) &= \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\left(\phi \geq \|\bar{x}_{\mathcal{G}_1} - \bar{x}_{\mathcal{G}_2}\| \,\middle|\, \mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}\left(\mathbf{x}'(\phi)\right)\right) \\
&= \frac{\mathbb{E}\left[\mathbb{1}\{\phi \geq \|\bar{x}_{\mathcal{G}_1} - \bar{x}_{\mathcal{G}_2}\|,\, \mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}\left(\mathbf{x}'(\phi)\right)\}\right]}{\mathbb{E}\left[\mathbb{1}\{\mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}\left(\mathbf{x}'(\phi)\right)\}\right]}
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Among all perturbations that preserve the clusters,
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Do I have room to push the clusters together while preserving the clustering ?

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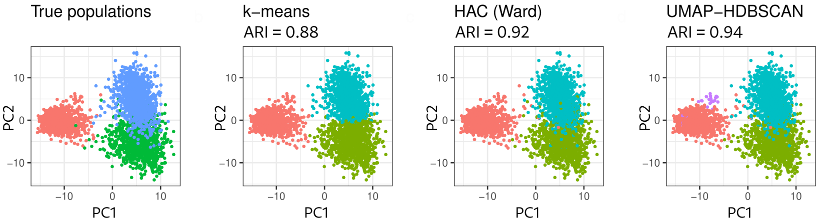
- If $\{\mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}(\mathbf{x}'(\phi))\}$ is unlikely, p-values will be very close to 0 or 1.

Consequences of conditioning (I)

Clustering need to be robust to small perturbations

Simulation setting

SNPs for $n = 3000$ individuals equally sampled from three populations[1] + Clustering performed on the first 100 PCs.



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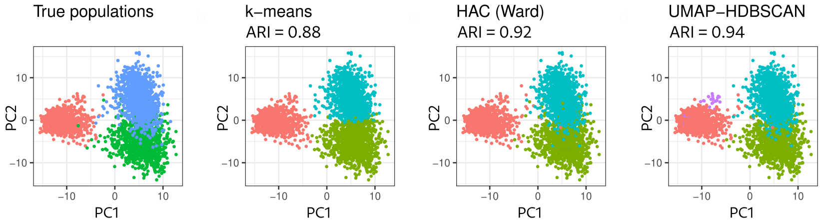

$\{\mathcal{G}_1, \mathcal{G}_2\}$	{●, ●}	{●, ●}	{●, ●}
k-means	0.279	0.554	0.178
HAC (Ward)	$< 10^{-16}$	$< 10^{-16}$	$< 10^{-16}$
UMAP+HDBSCAN	1	0.002	0.002

p-values obtained using `PCIdep` (González-Delgado *et al.* 2023).

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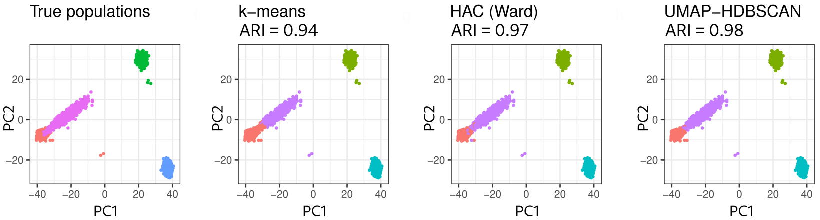
Consequences of conditioning (II)

Shapes and relative positions of clusters may hinder inference

Simulation setting

SNPs for $n = 4000$ individuals equally sampled from four populations : two sources, one unrelated population, and an admixed population derived from the sources[1] + Clustering performed on the first 100 PCs.



1. Data : Diaz-Papkovich, *unpublished*.

Consequences of conditioning (II)

Shapes and relative positions of clusters may hinder inference

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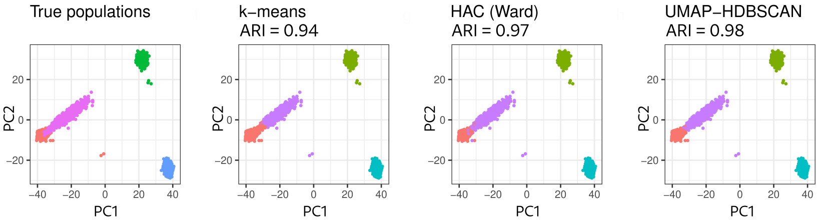

$\{\mathcal{G}_1, \mathcal{G}_2\}$	{●, ●}	{●, ●}	{●, ●}	{●, ●}	{●, ●}	{●, ●}
k-means	0.118	0.113	0.005	0.189	0.767	0.667
HAC	0.084	0.011	$< 10^{-16}$	0.343	$< 10^{-16}$	$< 10^{-16}$
UMAP+HDBSCAN	1	1	0.002	0.002	0.002	0.002

p-values obtained using `PCIdep` (González-Delgado *et al.* 2023).

1. Data : Diaz-Papkovich, *unpublished*.

Conclusion and perspectives

Main message

Conditional inference inherently limits post-clustering inference.

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The theory appears unsuitable in multiple practical applications

- Clustering algorithms sensitive to smooth boundaries,
- Realistic cluster placements.

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Limitations come from conditioning

Advances in this theory should proceed through a new selective inference framework, such as simultaneous inference :

$$\mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\left(\forall \{\mathcal{G}_1,\mathcal{G}_2\} : \text{Reject } H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}} \text{based on } \mathbf{X} \text{ at level } \alpha\right) \leq \alpha.$$

Ongoing exploration...

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Ongoing exploration...

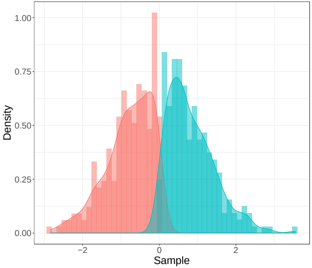
Thank you for your attention !

https://gonzalez-delgado.github.io/

Post-clustering inference

Toy example

- Simulate $\mathcal{N}(0,1) + \mathcal{U}(-0.2, 0.2)$
- Ask k-means to find 2 clusters (data-driven hypothesis selection)
- Test for the difference of cluster means (inference after selection)



- Ignoring adaptive selection : $p_Z = 10^{-67}$,
- Accounting for adaptive selection : $p_{AS} = 0.84$ (Chen and Witten 2023).

Adapted from Hivert *et al.* 2024

Post-clustering inference

Framework setting

- Let $C(\cdot)$ be a clustering algorithm, $\mathbf{X}$ a $n \times p$ random matrix with $\mathbb{E}(\mathbf{X}) = \boldsymbol{\mu}$.
- Let X_i (resp. μ_i) denote the i-th row of $\mathbf{X}$ (resp. $\boldsymbol{\mu}$) for $i \in [n] = \{1, \dots, n\}$.
- For any $\mathcal{G} \subset \{1, \dots, n\}$, let $\bar{X}_{\mathcal{G}} = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} X_i$ and $\bar{\mu}_{\mathcal{G}} = \frac{1}{|\mathcal{G}|} \sum_{i \in \mathcal{G}} \mu_i$.
- Let $\mathcal{G}_1, \mathcal{G}_2 \subset \{1, \dots, n\}$ be two non-overlapping groups of observations. Considering the column vector $\nu_{\mathcal{G}_1, \mathcal{G}_2} = \nu$ having as components

$$\nu_i = \mathbb{1}\{i \in \mathcal{G}_1\}/|\mathcal{G}_1| - \mathbb{1}\{i \in \mathcal{G}_2\}/|\mathcal{G}_2|,$$

  for $i \in [n]$, we can write the difference between the (empirical) group means as

$$\bar{\mu}_{\mathcal{G}_1} - \bar{\mu}_{\mathcal{G}_2} = \boldsymbol{\mu}^T \nu, \quad \text{and} \quad \bar{X}_{\mathcal{G}_1} - \bar{X}_{\mathcal{G}_2} = \mathbf{X}^T \nu.$$

We are interested in the following null hypothesis :

$$H_0^{\{\mathcal{G}_1, \mathcal{G}_2\}} : \boldsymbol{\mu}^T \nu = 0. \tag{H0}$$

Independence setting (I)

Gao, Bien and Witten 2022

Framework

Consider the model

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbf{I}_n, \sigma^2 \mathbf{I}_p), \qquad \text{(indep)}$$

that is, observations are independent and distributed as $X_i \sim \mathcal{N}_p(\mu_i, \sigma^2 \mathbf{I}_p)$.

p-value for (H0) under (indep) (Gao, Bien and Witten 2022)

$$p(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\Big(\|\mathbf{X}^T\nu\|_2 \geq \|\mathbf{x}^T\nu\|_2 \;\Big|\; \mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}(\mathbf{X}),$$
$$\boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x}, \, \mathrm{dir}\big(\mathbf{X}^T\nu\big) = \mathrm{dir}\big(\mathbf{x}^T\nu\big)\Big), \qquad \text{(p-GBW)}$$

where $\boldsymbol{\pi}_\nu^\perp = \mathbf{I}_n - \nu\nu^T/\|\nu\|_2^2$ and $\mathrm{dir}(v) = v/\|v\|_2 \mathbb{1}\{v \neq 0\}$ for all $v \in \mathbb{R}^p$.

The extra conditioning event allows to rewrite :

$$\{\mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}(\mathbf{X}), \boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x}, \, \mathrm{dir}\big(\mathbf{X}^T\nu\big) = \mathrm{dir}\big(\mathbf{x}^T\nu\big)\} \quad =$$
$$\{\|\mathbf{X}^T\nu\|_2 \in \mathcal{S}_2(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}), \, \boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x}, \, \mathrm{dir}\big(\mathbf{X}^T\nu\big) = \mathrm{dir}\big(\mathbf{x}^T\nu\big)\}.$$

Independence setting (II)

Gao, Bien and Witten 2022

p-value for (H0) under (indep) (Gao, Bien and Witten 2022)

$$p(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\Big(\|\mathbf{X}^T\nu\|_2 \geq \|\mathbf{x}^T\nu\|_2 \;\Big|\; \|\mathbf{X}^T\nu\|_2 \in \mathcal{S}_2(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}), \boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x}, \, \mathrm{dir}(\mathbf{X}^T\nu) = \mathrm{dir}(\mathbf{x}^T\nu)\Big).$$

Strategy to derive a tractable form of $p(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\})$

- $\mathbf{X}^T\nu \sim \mathcal{N}_p(0_p, \sigma^2\|\nu\|_2^2\, \mathbf{I}_p)$ under (H0),
- Choice of the norm $\|\cdot\|_2 \to \|\mathbf{X}^T\nu\|_2 \sim \sigma\|\nu\|_2 \cdot \chi_p$ under (H0),
- $\{\boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x}, \, \mathrm{dir}(\mathbf{X}^T\nu) = \mathrm{dir}(\mathbf{x}^T\nu)\}$ is independent of $\|\mathbf{X}^T\nu\|_2 \to$ $p(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\})$ is written as the CDF of a truncated χ_p.

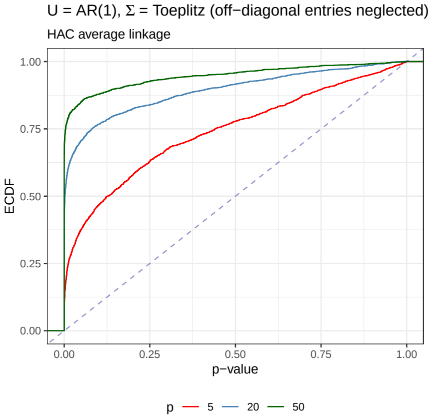
Theorem 1 in Gao, Bien and Witten 2022

$$p(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = 1 - \mathbb{F}_p\Big(\|\mathbf{X}^T\nu\|_2; \sigma\|\nu\|_2, \mathcal{S}_2(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) \underbrace{\boxed{\mathcal{S}_2(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\})}}_{\text{HAC, } k\text{-means}}\Big)$$

where $\mathbb{F}_p(t; c, \mathcal{S})$ denotes the CDF of a $c\chi_p$ random variable truncated to the set $\mathcal{S}$.

Ignoring dependency prevents selective type I error control

- Simulate $X_i \sim \mathcal{N}_p(\mathbf{0}_p, \boldsymbol{\Sigma})$ with $X_1, \ldots, X_n$ *dependent*.
- Set $\mathcal{C}$ to choose three clusters, randomly select two groups and test for the difference of their means assuming $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}_p$ and independent observations.



Arbitrary dependence setting

General strategy (I)

Framework

Consider the model

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbf{U}, \mathbf{\Sigma}) \Leftrightarrow \text{vec}(\mathbf{X}) \sim \mathcal{N}_{np}(\text{vec}(\boldsymbol{\mu}), \mathbf{\Sigma} \otimes \mathbf{U}), \qquad \text{(dep)}$$

where $\mathbf{U} \in \mathbb{R}^{n\times n}$ and $\mathbf{\Sigma} \in \mathbb{R}^{p\times p}$ are positive definite. That means :

- $X_i \sim \mathcal{N}_p(\mu_i, U_{ii}\mathbf{\Sigma})$ ($\mathbf{\Sigma} \leftrightarrow$ dependence between features),
- $X^j \sim \mathcal{N}_n(\mu^j, \Sigma_{jj}\mathbf{U})$ ($\mathbf{U} \leftrightarrow$ dependence between observations).

In this model, $\mathbf{X}^T\nu \sim \mathcal{N}_p(0_p, \mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2})$ under (H0), where $\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2} = \nu^T\mathbf{U}\nu\mathbf{\Sigma}$.
Therefore, $\|\mathbf{X}^T\nu\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}} \sim \chi_p$ under (H0), where $\|v\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}} = \sqrt{v^T\mathbf{V}^{-1}_{\mathcal{G}_1,\mathcal{G}_2}v}$, $v \in \mathbb{R}^p$.

Candidate p-value for (H0) under (dep)

$$p_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}; \{\mathcal{G}_1,\mathcal{G}_2\}) = \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\Big(\|\mathbf{X}^T\nu\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}} \geq \|\mathbf{x}^T\nu\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}} \;\Big|\; \mathcal{G}_1,\mathcal{G}_2 \in \mathcal{C}(\mathbf{X}), \boldsymbol{\pi}_\nu^\perp\mathbf{X} = \boldsymbol{\pi}_\nu^\perp\mathbf{x}, \, \mathrm{dir}_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{X}^T\nu) = \mathrm{dir}_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}^T\nu)\Big)$$

Arbitrary dependence setting

General strategy (II)

Candidate p-value for (H0) under (dep)

$$p_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x};\{\mathcal{G}_1,\mathcal{G}_2\}) = \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\Big(\|\mathbf{X}^T\nu\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}} \geq \|\mathbf{x}^T\nu\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}} \;\Big|\; \|\mathbf{X}^T\nu\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}} \in \mathcal{S}_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x};\{\mathcal{G}_1,\mathcal{G}_2\}),\; \boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x},\; \mathrm{dir}_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{X}^T\nu) = \mathrm{dir}_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}^T\nu)\Big).$$

For any $p \times p$ symmetric positive definite matrix $\mathbf{A}$, let $\|v\|_{\mathbf{A}}^2 = v^T\mathbf{A}^{-1}v$ and $\mathrm{dir}_{\mathbf{A}}(v) = v/\|v\|_{\mathbf{A}}\mathbb{1}\{v \neq 0\}$ for all $v \in \mathbb{R}^p$.

Proposition

(*i*) $\mathbf{A} = c\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}$ for some $c > 0 \overset{\text{(H0)}}{\Leftrightarrow} ||\mathbf{X}^T\nu_{\mathcal{G}_1,\mathcal{G}_2}||_{\mathbf{A}} \perp\!\!\!\perp \mathrm{dir}_{\mathbf{A}}\left(\mathbf{X}^T\nu_{\mathcal{G}_1,\mathcal{G}_2}\right)$,

(*ii*) $\mathbf{X}^T\nu_{\mathcal{G}_1,\mathcal{G}_2} \perp\!\!\!\perp \boldsymbol{\pi}^\perp_{\nu_{\mathcal{G}_1,\mathcal{G}_2}}\mathbf{X}$ for all $(\mathcal{G}_1,\mathcal{G}_2) \Leftrightarrow \mathbf{U} \in \mathcal{CS}(n)$,

where $\mathcal{CS}(n)$ is the class of compound symmetry positive definite matrices :

$$\mathcal{CS}(n) = \left\{(a-b)\mathbf{I}_n + b\mathbf{1}_{n\times n} \,:\, a \geq 0,\, -\frac{a}{n-1} < b < a\right\}.$$

Arbitrary dependence setting

Post-clustering inference for $\mathbf{U} \in \mathcal{CS}(n)$

The direct extension of the strategy of Gao *et al.* to the general model (dep) imposes a compound symmetry constraint on the dependence between observations.

Theorem

Let $\mathcal{C}$ be a clustering algorithm and $\mathbf{x}$ a realization of $\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbf{U}, \boldsymbol{\Sigma})$ with $\mathbf{U} \in \mathcal{CS}(n)$. Then,

$$p_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = 1 - \mathbb{F}_p\Big(\|\mathbf{x}^T \nu\|_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}, \mathcal{S}_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}, \{\mathcal{G}_1, \mathcal{G}_2\}) \underbrace{\boxed{\mathcal{S}_{\mathbf{V}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}, \{\mathcal{G}_1, \mathcal{G}_2\})}}_{\text{Scale trans. of } \mathcal{S}_2}\Big) \quad \text{(p-tract)}$$

where $\mathbb{F}_p(t, \mathcal{S})$ is the cumulative distribution function of a χ_p random variable truncated to the set $\mathcal{S}$.

- The control of the sel. type I error is robust to moderate deviations of $\mathbf{U} \in \mathcal{CS}(n)$.

What if $\mathbf{U} \notin \mathcal{CS}(n)$?

- The projection $\boldsymbol{\pi}_\nu^\perp \mathbf{X}$ is not independent of $\mathbf{X}^T\nu$ in general.
- Therefore, the distribution of interest is not that of $\mathbf{X}^T\nu$, but rather that of the conditioned vector :

$$\bar{\mathbf{X}}_\nu(\mathbf{x}) := \mathbf{X}^T\nu \,\big|\, \{\boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x},\, \mathrm{dir}(\mathbf{X}^T\nu) = \pm\mathrm{dir}(\mathbf{x}^T\nu)\}, \quad \text{for } \mathbf{x} \in \mathbb{R}^{n\times p},$$

Theorem

Let $\mathcal{C}$ be a clustering algorithm and $\mathbf{x}$ a realization of $\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbf{U}, \boldsymbol{\Sigma})$. Then,

$$\bar{\mathbf{X}}_\nu(\mathbf{x}) \sim \mathcal{N}_p\left(0, \boldsymbol{\Gamma}_\mathbf{x}\right),$$

under (H0), where

$$\boldsymbol{\Gamma}_\mathbf{x} = (\mathbf{I}_p \otimes \nu^T)(\boldsymbol{\Sigma} \otimes \mathbf{U} - (\boldsymbol{\Sigma} \otimes \mathbf{U})\mathbf{A}_\mathbf{x}{}^\top(\mathbf{A}_\mathbf{x}(\boldsymbol{\Sigma} \otimes \mathbf{U})\mathbf{A}_\mathbf{x}{}^\top)^\dagger \mathbf{A}_\mathbf{x}(\boldsymbol{\Sigma} \otimes \mathbf{U}))(\mathbf{I}_p \otimes \nu),$$

$$\mathbf{A}_\mathbf{x} = \begin{bmatrix} \boldsymbol{\pi}_{\mathbf{x}_\nu}^\perp (\mathbf{I}_p \otimes \boldsymbol{\pi}_\nu) \\ \mathbf{I}_p \otimes \boldsymbol{\pi}_\nu^\perp \end{bmatrix},$$

$$\boldsymbol{\pi}_\nu = \mathbf{I}_n - \boldsymbol{\pi}_\nu^\perp,\ \mathbf{x}_\nu = \mathrm{vec}(\boldsymbol{\pi}_\nu \mathbf{x}) \quad \text{and} \quad \boldsymbol{\pi}_{\mathbf{x}_\nu}^\perp = \mathbf{I}_{np} - \mathbf{x}_\nu^T \mathbf{x}_\nu / \|\mathbf{x}_\nu\|_2^2.$$

Arbitrary $\mathbf{U}$ structures

Candidate p-value for (H0) under arbitrary $\mathbf{U}$

$$p_{\mathbf{\Gamma}}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = \mathbb{P}_{H_0^{\{\mathcal{G}_1,\mathcal{G}_2\}}}\Big(\|\mathbf{X}^T\nu\|_{\mathbf{\Gamma}_\mathbf{x}} \geq \|\mathbf{x}^T\nu\|_{\mathbf{\Gamma}_\mathbf{x}} \;\Big|\; \mathcal{G}_1, \mathcal{G}_2 \in \mathcal{C}(\mathbf{X}),$$
$$\boldsymbol{\pi}_\nu^\perp \mathbf{X} = \boldsymbol{\pi}_\nu^\perp \mathbf{x}\,,\, \mathrm{dir}(\mathbf{X}^T\nu) = \pm\mathrm{dir}(\mathbf{x}^T\nu) \Big),$$

where $\|v\|_{\mathbf{\Gamma}_\mathbf{x}}^2 = v^T \mathbf{\Gamma}_\mathbf{x}^\dagger v, \quad \forall\, v \in \mathbb{R}^p$.

Proposition

The quantity $\|\bar{\mathbf{X}}_\nu(\mathbf{x})\|_{\mathbf{\Gamma}_\mathbf{x}}$ follows $\mathbf{x}$-a.s. a χ_1 distribution under (H0). Moreover,

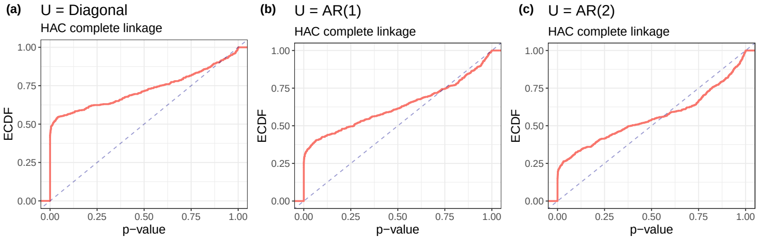
$$p_{\mathbf{\Gamma}}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = 1 - \mathbb{F}_1\left(\|\mathbf{x}^T\nu\|_{\mathbf{\Gamma}_\mathbf{x}}, \mathcal{S}_{\mathbf{\Gamma}_\mathbf{x}}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) \right),$$

where $\mathbb{F}_1(t, \mathcal{S})$ is the cumulative distribution function of a χ_1 random variable truncated to the set $\mathcal{S}$ and $\mathcal{S}_{\mathbf{\Gamma}_\mathbf{x}}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\})$ is a scale transformation of $\mathcal{S}_2$.

- Assessing whether $p_{\mathbf{\Gamma}}(\mathbf{X}; \{\mathcal{G}_1, \mathcal{G}_2\})$ controls the selective type I error is a challenging problem, as it requires understanding the behavior of the null distribution of $\|\bar{\mathbf{X}}_\nu(\mathbf{x})\|_{\mathbf{\Gamma}_\mathbf{x}}^2 = \bar{\mathbf{X}}_\nu(\mathbf{x})^T \mathbf{\Gamma}_\mathbf{x}^\dagger \bar{\mathbf{X}}_\nu(\mathbf{x})$.

Arbitrary $\mathbf{U}$ structures

Numerical simulations suggest the unsuitability of $p_{\boldsymbol{\Gamma}}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\})$



Conclusion

Defining a tractable p-value that ensures the selective type I error control requires the conditioning on events that are *independent* of the test statistic.

Estimation of unknown parameters

Independence setting (Gao *et al.* 2022)

Let $\mathbf{X}^{(n)} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}^{(n)}, \mathbf{I}_n, \sigma^2\mathbf{I}_p)$ and consider

$$\hat{p}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = 1 - \mathbb{F}_p\Big(\|\mathbf{x}^T\nu\|_2; \hat{\sigma}\|\nu\|_2, \mathcal{S}_2(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\})\Big).$$

Theorem 4 in Gao *et al.* 2022

If $\hat{\sigma}$ is an estimator of σ such that

$$\lim_{n\to\infty} \mathbb{P}_{H_0^{\{\mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)}\}}}\left(\hat{\sigma}\left(\mathbf{X}^{(n)}\right) \geq \sigma \,\middle|\, \mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)} \in \mathcal{C}\left(\mathbf{X}^{(n)}\right)\right) = 1, \qquad (\sigma \text{ over-est})$$

then, for any $\alpha \in [0, 1]$, it holds

$$\limsup_{n\to\infty} \mathbb{P}_{H_0^{\{\mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)}\}}}\left(\hat{p}\left(\mathbf{X}^{(n)}; \{\mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)}\}\right) \leq \alpha \,\middle|\, \mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)} \in \mathcal{C}\left(\mathbf{X}^{(n)}\right)\right) \leq \alpha.$$

$\rightarrow$ Gao *et al.* propose an estimator $\hat{\sigma}$ that satisfies (σ over-est) under mild assumptions on $\{\boldsymbol{\mu}^{(n)}\}_{n\in\mathbb{N}}$.

Estimation of unknown parameters

Arbitrary dependence setting

Let

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbf{U}, \boldsymbol{\Sigma}). \qquad \text{(dep)}$$

Can we estimate both $\mathbf{U}$ and $\boldsymbol{\Sigma}$?

- Under the general model (dep), the scale matrices $\mathbf{U}$ and $\boldsymbol{\Sigma}$ are non-identifiable.
- Multiple copies of $\mathbf{X}$ are needed to simultaneously estimate $\mathbf{U}$ and $\boldsymbol{\Sigma}$.
- We assume that either $\mathbf{U}$ or $\boldsymbol{\Sigma}$ is known :

$$\mathbf{X} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}, \mathbf{U}, \boldsymbol{\Sigma}) \Leftrightarrow \mathbf{X}^T \sim \mathcal{MN}_{p\times n}(\boldsymbol{\mu}^T, \boldsymbol{\Sigma}, \mathbf{U}).$$

$\rightarrow$ How to extend the notion of over-estimation to matrices ?

Loewner partial order $\succeq$

Let A, B be two Hermitian matrices. $A \succeq B$ if and only if $A - B$ is positive semidefinite (PSD).

Remark : If $A = \hat{\sigma}\mathbf{I}_p$ and $B = \sigma\mathbf{I}_p$, the condition $A \succeq B$ becomes $\hat{\sigma} \geq \sigma$.

Over-estimation of $\boldsymbol{\Sigma}$ for known $\mathbf{U}$

Let $\mathbf{X}^{(n)} \sim \mathcal{MN}_{n\times p}(\boldsymbol{\mu}^{(n)}, \mathbf{U}^{(n)}, \boldsymbol{\Sigma})$ with $\mathbf{U}^{(n)} \in \mathcal{CS}(n)$ and consider

$$p_{\hat{\mathbf{V}}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}; \{\mathcal{G}_1, \mathcal{G}_2\}) = 1 - \mathbb{F}_p\left(\|\mathbf{x}^T \nu\|_{\hat{\mathbf{V}}_{\mathcal{G}_1,\mathcal{G}_2}}; \mathcal{S}_{\hat{\mathbf{V}}_{\mathcal{G}_1,\mathcal{G}_2}}(\mathbf{x}, \{\mathcal{G}_1, \mathcal{G}_2\}) \right)$$

where $\hat{\mathbf{V}}_{\mathcal{G}_1,\mathcal{G}_2} = \nu^T \mathbf{U} \nu \hat{\boldsymbol{\Sigma}}(\mathbf{x})$.

Theorem (extension of Theorem 4 in Gao *et al.* 2022)

Let $\mathbf{U} \in \mathcal{CS}(n)$ and $\hat{\boldsymbol{\Sigma}}\left(\mathbf{X}^{(n)}\right)$ be a positive definite estimator of $\boldsymbol{\Sigma}$ such that

$$\lim_{n\to\infty} \mathbb{P}_{H_0^{\{\mathcal{G}_1^{(n)},\mathcal{G}_2^{(n)}\}}}\left(\hat{\boldsymbol{\Sigma}}(\mathbf{X}^{(n)}) \succeq \boldsymbol{\Sigma} \,\middle|\, \mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)} \in \mathcal{C}\left(\mathbf{X}^{(n)}\right) \right) = 1, \qquad (\boldsymbol{\Sigma} \text{ over-est})$$

then, for any $\alpha \in [0, 1]$, we have

$$\limsup_{n\to\infty} \mathbb{P}_{H_0^{\{\mathcal{G}_1^{(n)},\mathcal{G}_2^{(n)}\}}}\left(p_{\hat{\mathbf{V}}_{\mathcal{G}_1^{(n)},\mathcal{G}_2^{(n)}}}(\mathbf{X}^{(n)}; \{\mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)}\}) \le \alpha \,\middle|\, \mathcal{G}_1^{(n)}, \mathcal{G}_2^{(n)} \in \mathcal{C}\left(\mathbf{X}^{(n)}\right) \right) \le \alpha.$$

$\rightarrow$ We propose an estimator $\hat{\boldsymbol{\Sigma}}$ that satisfies ($\boldsymbol{\Sigma}$ over-est) under mild assumptions on $\{\boldsymbol{\mu}^{(n)}\}$ and for several common models of dependence $\{\mathbf{U}^{(n)}\}$.

- Preprint : https://arxiv.org/abs/2310.11822,
- R package `PCIdep` at https://github.com/gonzalez-delgado/PCIdep/.

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